

CASE STUDY 01 — CIVIC TECH

SEVA SETU

Digital Civic Complaint Management Platform



Developer	Madhura Surve
Role	Backend Architect & Relational Schema Designer
Framework	FastAPI (Python) + Supabase + React Native
Status	Nearing Production Launch (Local Municipal Trials)
Source Code	github.com/madhurasurve708-prog/seva-setu

1. Executive Summary

Seva Setu is a centralized complaint management platform designed to bridge the communication gap between local citizens and municipal administrators. By building a high-performance backend, modeling normalized relational schemas, and implementing seamless authentication, this system transforms messy civic processes into clear, traceable, and structured digital workflows.

2. The Challenge & Core Research

Traditional civic complaint handling involves scattered phone calls, physical logs, and a lack of ticket transparency. Citizens have no way to verify if their concerns (such as road repair, waste management, or water issues) are being addressed, while municipal staff struggle with manual routing and division tracking.

Field Research Highlight: To ground the system in actual workflows, user surveys were conducted with local citizens to map out resolution friction. Additionally, we collaborated directly with municipal officials to align the administrative features with actual city division operations and escalation rules.

3. System Architecture & Relational Design

I designed and developed the FastAPI backend application in Python, utilizing PostgreSQL as the core relational database. The architecture is organized to enforce strict role-based access control (citizens, department representatives, ward officers, and administrators). For secure object storage, I integrated Supabase buckets to handle citizen photo evidence uploads directly upon complaint creation.

4. Developer Role & Contributions

- **Relational Schemas:** Modeled and implemented database schemas mapping complex associations between citizens, complaints, departments, wards, and escalation histories.
- **REST API Backend:** Built FastAPI service controllers, routing logic, validation layers, and custom payload handlers.
- **Supabase Storage integration:** Programmed backend pipelines to secure file paths and upload citizen complaint photos.
- **Stakeholder Alignment:** Transformed municipal administrative protocols into programmatic rules, enabling ward-based auto-routing.

5. Featured Technology Stack

Layer	Technologies	Core Purpose / Application
Backend / REST API	Python, FastAPI	High-speed routing, validation (Pydantic), dependency injection, and API endpoint implementation.
Database / Storage	PostgreSQL, Supabase	Relational schema storage, data normalization, transaction isolation, and secure bucket storage for media uploads.
Mobile Front-End	React Native, Expo, TS	Cross-platform mobile application utilizing Expo modules to capture images, location data, and track ticket status (built by partner).
Security / Access	JWT, Supabase Auth	Role-based token validation (RBAC) ensuring secure separation between citizen features and municipal dashboards.

6. Key Platform Capabilities

The backend application implements several key functional workflows:

- **Citizen Authentication & Session Management:** Secure login mechanisms for citizens to manage their ticket registry.
- **Interactive Complaint Registry:** API pathways enabling complaints to be filed with descriptions, department tags, and photo attachments.
- **Automated Ward-Based Routing:** Logic mapping complaint coordinates to specific ward jurisdiction tables, sending alerts to corresponding ward officials.
- **Official Dashboards & Escalation:** Web pathways for department representatives to accept tickets, post updates, or escalate to senior officers.
- **Real-Time Progress Updates:** Citizen tracking routes mapping ticket lifecycle status (Submitted → In Progress → Resolved).

7. Project Results & Current Status

Current Status: Seva Setu has successfully transitioned from an initial concept prototype to a field-vetted digital platform. The REST API backend has been fully tested, and the integrated mobile application is currently nearing its official production launch, scheduled for trial runs in local municipal wards.